

Yash Pratap Singh

Computer Science Engineering Student

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PROFESSIONAL SUMMARY

B.Tech Computer Science student (Class of 2028) with hands-on experience in Java, C++, and Python, and a working knowledge of data structures, algorithms, object-oriented design, and database management systems. Built and presented a multithreaded C++ server handling concurrent requests at scale, and developed a database-driven academic scheduling system. Quick to learn new technologies, comfortable with ambiguity, and experienced contributing to automated testing and system design in team settings.

TECHNICAL SKILLS

Programming Languages: Java, C++, C, Python, Swift

Web Technologies: HTML, CSS, JavaScript (familiar with React, Angular, Vue.js fundamentals)

Databases: SQL, MySQL, Relational Database Design

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Multithreading, Distributed Systems, Algorithm Complexity Analysis

Tools & Platforms: Git, IntelliJ, SwiftUI, Socket Programming, TCP/IP

LEADERSHIP & EXTRACURRICULAR INVOLVEMENT

President, Coding Club — SRM Institute of Science and Technology (SRMIST)

2025 – Present

- Appointed to lead the institute's Coding Club through a competitive, merit-based selection process conducted by the Head of Department, reflecting strong technical credibility and demonstrated leadership potential.

PROJECTS

Compact Multithreaded Web Server for Static Request Handling

2025

C++ | Socket Programming | Multithreading | Thread Pool | TCP

- Developed a lightweight multithreaded HTTP web server in C++ using POSIX sockets to handle concurrent static content requests
- Implemented a priority-based thread pool and load-balanced task queue to improve request scheduling and overall server throughput
- Presented research paper on the server's architecture at ICAECT 2026, an IEEE-sponsored international conference, demonstrating efficient concurrent HTTP request handling

Intelligent Academic ERP System

2026

DBMS | Algorithm Design | ERP Architecture

- Designed an academic ERP prototype integrating relational database design with conflict-free scheduling algorithms
- Applied machine learning models to predict resource utilization and support data-driven academic planning

EDUCATION

Bachelor of Technology in Computer Science, SRM Institute of Science and Technology (SRMIST), Trichy, India 07/2024 – 07/2028

- Relevant coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

Senior School Certificate (SSC), Vatayan School, Bihar

2021 – 2023

Indian Certificate of Secondary Education, Narayana School

2016 – 2021

ACHIEVEMENTS

- Solved 150+ data structures and algorithms problems on LeetCode
- Published and presented a research paper, "Compact Multithreaded Web Server for Static Request Handling," at ICAECT 2026, an IEEE-sponsored international conference